



Assignments I, and II

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1 Assignment I

Instructions. All steps are shown. Notation: vectors are columns unless a row is written for compactness; $\|\cdot\|$ is the Euclidean (ℓ_2) norm; inner products are written $\langle x, y \rangle$ or $x^\top y$.

Question 1: Introduction to Machine Learning

(a) Definition. Machine learning is the study of algorithms that improve a well-specified performance measure by extracting statistical regularities from data, rather than by being given an exhaustive set of hand-written rules. Formally, a learner is given a hypothesis class $\mathcal{H}$ and a finite sample drawn from an unknown distribution $\mathcal{D}$, and it returns $h \in \mathcal{H}$ whose expected risk under $\mathcal{D}$ is small. The defining property is *generalisation*: the hypothesis must perform well on previously unseen draws from the same process.

(b) Supervised versus unsupervised learning. The distinction is whether the learning objective is defined with respect to an observed target.

- **Supervised learning.** Each training pair is (x_i, y_i) . The goal is to estimate $f : \mathcal{X} \rightarrow \mathcal{Y}$ that minimises a loss $\ell(f(x), y)$. Discrete $\mathcal{Y}$ yields classification; continuous $\mathcal{Y}$ yields regression.
- *Example.* Diagnosis from an electronic health record: laboratory panels and device telemetry are features; the clinician-assigned condition is the label. Logistic regression, a decision tree, an SVM, or k -NN all estimate $P(Y | X)$.
- **Unsupervised learning.** Only $\{x_i\}$ is observed. The objective is an internal criterion: a partition (clustering), a low-dimensional embedding (PCA), a density (Gaussian mixtures), or association rules. Evaluation is necessarily indirect.
- *Example.* Grouping unlabelled network flows into behavioural clusters, or compressing a sensor stream onto the leading principal components.

Supervision is the presence of a target in the loss, not the mere existence of structure in X .

(c) Why data are indispensable. A learning algorithm is an estimator. The Bayes predictor $f^*(x) = \arg \min_a \mathbb{E}[\ell(a, Y) | X = x]$ is a functional of $\mathcal{D}$ and is identified only through draws from $\mathcal{D}$. Uniform convergence bounds imply that the gap between empirical and true risk scales with the complexity of $\mathcal{H}$ and inversely with $\sqrt{n}$: capacity must be paid for in samples. Features, encodings and metrics that separate classes in one domain fail in another. The model is a vessel; the data are the content.

Question 2: Measurement Scales and Attributes

Stevens' typology: an *interval* scale has a meaningful unit and therefore meaningful differences, but an arbitrary origin. A *ratio* scale additionally possesses a true zero and therefore admits multiplication and division.

Variable	Before	After
IQ score	80	160
Annual funding	\$50,000	\$100,000
Execution time	2 hours	4 hours

(a) **“The researcher became twice as intelligent.” — Invalid.** Deviation IQ is constructed with a conventional mean of 100 and SD of 15. Zero is not the absence of intelligence, so ratio statements of the form “twice as intelligent” have no empirical content. One may report an 80-point increase or a change of percentile rank, but not a doubling of intelligence.

(b) **“The annual funding doubled.” — Valid.** Currency is a ratio attribute: zero dollars denotes the absence of funding, and

$$\frac{100000}{50000} = 2.$$

(c) **“The algorithm required twice as much execution time.” — Valid.** Elapsed time is a duration with a true zero (no time has passed). Clock-of-day would be interval; that is not what is measured here.

(d) **“The increase in IQ score was 80 points.” — Valid.** Interval scales support differences: $160 - 80 = 80$. Differences are licensed; ratios of levels are not.

(e) **Classification.**

Variable	Scale	Justification
IQ score	Interval	Equal units; origin is conventional.
Annual funding	Ratio	True zero (no funds).
Execution time	Ratio	Duration from a true zero.

(f) **Significance of a true zero.** A true zero is the point at which the measured attribute is absent. On a ratio scale the numerical value is proportional to the underlying magnitude, so “ x is twice y ” is invariant under a change of unit. On an interval scale the origin may be shifted by an arbitrary c , and $(x + c)/(y + c)$ is not invariant. Before reporting a fold-increase or a coefficient of variation, confirm that the attribute has a true zero.

Question 3: Data Representation

The design matrix is

$$X = \begin{bmatrix} 2 & 3 \\ 4 & 1 \\ 5 & 6 \\ 1 & 8 \end{bmatrix} \in \mathbb{R}^{4 \times 2}.$$

(a) $n = 4$ samples, $d = 2$ features.

(b) The third sample as a column is

$$x_3 = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

(c) Second-feature coordinates: 3, 1, 6, 8. The unique maximum is 8, attained by the fourth sample $x_4 = (1, 8)$.

(d) In the plane whose axes are the two features, $(5, 6)$ is the point five units along the first axis and six along the second. Its distance from the origin is $\|x_3\|_2 = \sqrt{61}$.

Question 4: Vector Operations and Distance

Let $x_1 = (2, 3)$ and $x_3 = (5, 6)$.

(a)

$$x_3 - x_1 = (5 - 2, 6 - 3) = (3, 3).$$

(b)

$$d_2(x_1, x_3) = \|x_3 - x_1\|_2 = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}.$$

(c)

$$m = \frac{x_1 + x_3}{2} = (3.5, 4.5).$$

(d) Euclidean distance is the default geometry of $\mathbb{R}^d$: rotationally invariant, induced by an inner product, and compatible with the isotropic Gaussian. It is the metric of k -NN, the reconstruction loss of k -means, and the geometry recovered by a kernel $k(x, z) = \langle \phi(x), \phi(z) \rangle$. A caveat: the metric treats coordinates as commensurate. If units differ, standardise or replace d_2 by a Mahalanobis metric.

Question 5: Euclidean Norm and Unit Vectors

(a)

$$\begin{aligned} \|(-3, 4)\|_2 &= \sqrt{9 + 16} = 5, & \hat{u} &= (-3/5, 4/5), \\ \|(1, 2, 2)\|_2 &= \sqrt{1 + 4 + 4} = 3, & \hat{u} &= (1/3, 2/3, 2/3), \\ \|(1, 1, 1)\|_2 &= \sqrt{4} = 2, & \hat{u} &= (1/2, 1/2, 1/2). \end{aligned}$$

Each unit vector has norm 1 by direct substitution.

(b) A unit vector is a pure direction: positive scaling of v leaves $\hat{u} = v/\|v\|$ unchanged. Cosine similarity is the inner product of two unit vectors. Gradient steps are often taken in the direction $-\nabla L/\|\nabla L\|$ so that the step-size is not confounded with raw gradient magnitude. Whenever magnitude encodes an uninteresting size effect (document length, illumination), replacing v by $\hat{u}$ removes that confound.

Question 6: Cosine Similarity

Let $x = (1, 2, 3)$ and $y = (2, 4, 6)$.

(a)

$$\begin{aligned}\langle x, y \rangle &= 2 + 8 + 18 = 28, & \|x\|_2 &= \sqrt{14}, & \|y\|_2 &= \sqrt{56} = 2\sqrt{14}, \\ \cos \theta &= \frac{28}{\sqrt{14} \cdot 2\sqrt{14}} = \frac{28}{28} = 1.\end{aligned}$$

(b) A cosine of 1 means $\theta = 0$: the vectors are positive scalar multiples ($y = 2x$) and represent the same pattern up to scale. Cosine is blind to magnitude, which is intended when length is a nuisance and a defect when length is signal.

(c) In information retrieval a query and a document are encoded as TF-IDF (or embedding) vectors. Cosine is the ranking score, preferred to Euclidean distance because two documents on the same topic at different lengths would otherwise be driven apart by the extra mass of the longer document. The same construction appears in collaborative filtering and as the argument of the softmax in contrastive losses.

2 Assignment II

Vector projection, eigenvalues, and probability distributions.

Instructions. All algebraic steps are shown and each numerical claim is justified. Lecture-note asides have been removed; only the argument required by the brief is retained.

Question 1

Given

$$\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 4 \end{pmatrix},$$

find the projection of $\mathbf{b}$ onto $\mathbf{a}$, the perpendicular component, and verify $\mathbf{b} = \mathbf{b}_{\parallel\mathbf{a}} + \mathbf{b}_{\perp\mathbf{a}}$.

Projection. The orthogonal projection of $\mathbf{b}$ onto $\text{span}\{\mathbf{a}\}$ is the unique vector $\mathbf{b}_{\parallel\mathbf{a}} = \alpha\mathbf{a}$ whose residual is orthogonal to $\mathbf{a}$:

$$\mathbf{a}^\top(\mathbf{b} - \alpha\mathbf{a}) = 0 \implies \alpha = \frac{\mathbf{a}^\top\mathbf{b}}{\mathbf{a}^\top\mathbf{a}}.$$

$$\mathbf{a}^\top\mathbf{b} = 2 \cdot 3 + 1 \cdot 4 = 10, \quad \mathbf{a}^\top\mathbf{a} = 4 + 1 = 5, \quad \alpha = \frac{10}{5} = 2.$$

$$\mathbf{b}_{\parallel\mathbf{a}} = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

Perpendicular component.

$$\mathbf{b}_{\perp\mathbf{a}} = \mathbf{b} - \mathbf{b}_{\parallel\mathbf{a}} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

Verification.

$$\mathbf{b}_{\parallel\mathbf{a}} + \mathbf{b}_{\perp\mathbf{a}} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \mathbf{b}.$$

Orthogonality: $\mathbf{a}^\top\mathbf{b}_{\perp\mathbf{a}} = 2(-1) + 1 \cdot 2 = 0$.

Geometrically $\mathbf{b}_{\parallel\mathbf{a}}$ is the closest point of $\text{span}\{\mathbf{a}\}$ to $\mathbf{b}$. This construction is the one-dimensional case of least-squares regression and of the Rayleigh argument that underlies PCA.

Question 2

Given

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

Projection.

$$\mathbf{a}^\top \mathbf{b} = 2 + 0 + 2 = 4, \quad \mathbf{a}^\top \mathbf{a} = 1 + 4 + 4 = 9,$$

$$\mathbf{b}_{\parallel \mathbf{a}} = \frac{4}{9} \mathbf{a} = \begin{pmatrix} 4/9 \\ 8/9 \\ 8/9 \end{pmatrix}.$$

Angle.

$$\|\mathbf{a}\| = 3, \quad \|\mathbf{b}\| = \sqrt{5}, \quad \cos \theta = \frac{4}{3\sqrt{5}}, \quad \theta = \cos^{-1}\left(\frac{4}{3\sqrt{5}}\right).$$

Numerically $\cos \theta \approx 0.5963$, so $\theta \approx 53.4^\circ$.

Residual and orthogonality.

$$\mathbf{b}_{\perp \mathbf{a}} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 4/9 \\ 8/9 \\ 8/9 \end{pmatrix} = \begin{pmatrix} 14/9 \\ -8/9 \\ 1/9 \end{pmatrix}.$$

$$\mathbf{a}^\top \mathbf{b}_{\perp \mathbf{a}} = \frac{14}{9} - \frac{16}{9} + \frac{2}{9} = 0.$$

Decomposition check:

$$\begin{pmatrix} 4/9 \\ 8/9 \\ 8/9 \end{pmatrix} + \begin{pmatrix} 14/9 \\ -8/9 \\ 1/9 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \mathbf{b}.$$

Question 3

Show that the projection matrix onto a nonzero vector $\mathbf{a}$ is

$$\mathbf{P} = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}}$$

and prove $\mathbf{P}^2 = \mathbf{P}$.

Derivation.

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} \mathbf{a} = \mathbf{a} \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{a}^\top \mathbf{a}} = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}} \mathbf{b}.$$

The linear map $\mathbf{b} \mapsto \text{proj}_{\mathbf{a}} \mathbf{b}$ is therefore represented by

$$\mathbf{P} = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}}.$$

Idempotence.

$$\mathbf{P}^2 = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}} \cdot \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}} = \frac{\mathbf{a}(\mathbf{a}^\top \mathbf{a})\mathbf{a}^\top}{(\mathbf{a}^\top \mathbf{a})^2} = \frac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top \mathbf{a}} = \mathbf{P},$$

where $\mathbf{a}^\top \mathbf{a}$ was cancelled as a positive scalar. Geometrically, a vector already on $\text{span}\{\mathbf{a}\}$ is left unchanged by a second projection.

Further properties used later. $\mathbf{P}$ is symmetric ($\mathbf{P}^\top = \mathbf{P}$); its eigenvalues are 1 (multiplicity one, eigenspace $\text{span}\{\mathbf{a}\}$) and 0 (multiplicity $n - 1$, eigenspace $\mathbf{a}^\perp$).

Question 4

Show that projection minimises $\|\mathbf{b} - \alpha\mathbf{a}\|^2$ and find the optimal α .

Let $f(\alpha) = \|\mathbf{b} - \alpha\mathbf{a}\|^2 = (\mathbf{b} - \alpha\mathbf{a})^\top(\mathbf{b} - \alpha\mathbf{a})$. Expanding,

$$f(\alpha) = \mathbf{b}^\top\mathbf{b} - 2\alpha(\mathbf{a}^\top\mathbf{b}) + \alpha^2(\mathbf{a}^\top\mathbf{a}).$$

Stationarity:

$$f'(\alpha) = -2\mathbf{a}^\top\mathbf{b} + 2\alpha\mathbf{a}^\top\mathbf{a} = 0 \implies \alpha^* = \frac{\mathbf{a}^\top\mathbf{b}}{\mathbf{a}^\top\mathbf{a}}.$$

The second derivative $f''(\alpha) = 2\|\mathbf{a}\|^2 > 0$ (since $\mathbf{a} \neq \mathbf{0}$) confirms a minimum. Equivalently, the normal equation $\mathbf{a}^\top(\mathbf{b} - \alpha\mathbf{a}) = 0$ is the orthogonality condition of Questions 1–3. Substituting α^* yields the minimal residual

$$\min_{\alpha} \|\mathbf{b} - \alpha\mathbf{a}\|^2 = \|\mathbf{b}\|^2 - \frac{(\mathbf{a}^\top\mathbf{b})^2}{\mathbf{a}^\top\mathbf{a}}.$$

For the vectors of Question 2 one recovers $\alpha^* = 4/9$.

Question 5

Find the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

and interpret them geometrically.

Characteristic equation.

$$\det(\mathbf{A} - \lambda I) = (2 - \lambda)^2 - 1 = 0 \implies 2 - \lambda = \pm 1 \implies \lambda_1 = 3, \quad \lambda_2 = 1.$$

Eigenvectors. For $\lambda = 3$: $(\mathbf{A} - 3I)\mathbf{v} = 0$ gives $-x + y = 0$, so $\mathbf{v}_1 = (1, 1)^\top$. For $\lambda = 1$: $(\mathbf{A} - I)\mathbf{v} = 0$ gives $x + y = 0$, so $\mathbf{v}_2 = (1, -1)^\top$. Check:

$$\mathbf{A} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\mathbf{v}_1, \quad \mathbf{A} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\mathbf{v}_2.$$

Geometry. $\mathbf{A}$ stretches the line $y = x$ by 3 and leaves the orthogonal line $y = -x$ unchanged. Because $\mathbf{A}$ is symmetric the eigenbasis is orthogonal, and

$$\mathbf{A} = PDP^{-1}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \text{diag}(3, 1).$$

Both eigenvalues are positive, so $\mathbf{A}$ is positive definite; $\text{tr}(\mathbf{A}) = 4 = \lambda_1 + \lambda_2$ and $\det(\mathbf{A}) = 3 = \lambda_1\lambda_2$.

Question 6

For

$$\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

compute the eigenvalues and explain how $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ scales space.

$\mathbf{A}$ is already diagonal, so $\lambda_1 = 3$ with $\mathbf{v}_1 = (1, 0)^\top$ and $\lambda_2 = 1$ with $\mathbf{v}_2 = (0, 1)^\top$. For $\mathbf{x} = (x, y)^\top$,

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = 3x^2 + y^2.$$

On the unit circle the quadratic form ranges from 1 (along e_2) to 3 (along e_1). Level sets $\mathbf{x}^\top \mathbf{A} \mathbf{x} = c$ are axis-aligned ellipses

$$\frac{x^2}{c/3} + \frac{y^2}{c} = 1,$$

compressed along x because that direction is scaled more strongly. In PCA language, $\mathbf{A}$ is a covariance that has already been written in its eigenbasis.

Question 7

Explain why eigenvectors represent principal directions in data.

Let Σ be the covariance of a centred sample. The variance of the one-dimensional projection onto a unit vector u is the Rayleigh quotient

$$\text{Var}(u^\top x) = u^\top \Sigma u.$$

A symmetric matrix attains its largest Rayleigh value at the leading eigenvector, and the value attained is the leading eigenvalue:

$$\max_{\|u\|=1} u^\top \Sigma u = \lambda_{\max}(\Sigma), \quad u^* = v_{\max}.$$

The next eigenvector solves the same problem in the orthogonal complement, and so on. Consequently:

- the eigenvectors of Σ are the orthogonal directions of extremal variance;
- the associated eigenvalues are those variances;
- coordinates in the eigenbasis are uncorrelated (spectral theorem: $V^\top \Sigma V = \Lambda$).

Keeping the top k eigenvectors is PCA: most of the trace of Σ is retained in k uncorrelated coordinates.

Question 8

Define the Bernoulli distribution and derive its mean and variance.

$X \sim \text{Bernoulli}(p)$ for $p \in [0, 1]$ if $X \in \{0, 1\}$ with $P(X = 1) = p$. Equivalently,

$$P(X = x) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

$$\mathbb{E}[X] = 0 \cdot (1 - p) + 1 \cdot p = p.$$

$$\mathbb{E}[X^2] = 0^2 \cdot (1 - p) + 1^2 \cdot p = p, \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

The variance is maximised at $p = 1/2$ ($\text{Var} = 1/4$) and vanishes at the deterministic endpoints $p \in \{0, 1\}$.

Question 9

For $X \sim \text{Bernoulli}(0.3)$, compute the mean and variance and interpret them in classification.

$$\mathbb{E}[X] = 0.3, \quad \text{Var}(X) = 0.3 \cdot 0.7 = 0.21.$$

If X is a class indicator ($1 = \text{positive}$), the mean is the prior $P(Y = 1) = 0.3$: thirty percent of instances are positive. A constant classifier that always predicts the majority class therefore attains accuracy 0.7; any reported accuracy must be read against that baseline. The variance 0.21 is close to the Bernoulli maximum 0.25, so the label is moderately uncertain. Class imbalance of this size is why precision, recall and macro- F_1 are preferred to raw accuracy, and why a decision threshold of $1/2$ need not be optimal for logistic regression (whose likelihood is exactly Bernoulli).

Question 10

Show that Bernoulli is the $n = 1$ case of the Binomial.

The Binomial PMF is $P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}$ for $y = 0, \dots, n$. Setting $n = 1$,

$$P(Y = 0) = \binom{1}{0} p^0 (1-p)^1 = 1-p, \quad P(Y = 1) = \binom{1}{1} p^1 (1-p)^0 = p,$$

which is the Bernoulli PMF. Means, variances and moment-generating functions coincide: $\mathbb{E} = p$, $\text{Var} = p(1-p)$, $\text{MGF} = 1-p+pe^t$. Thus $\text{Bernoulli}(p) \equiv \text{Binomial}(1, p)$.

Question 11

Derive $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ from Bernoulli trials.

Let $X_1, \dots, X_n$ be i.i.d. $\text{Bernoulli}(p)$ and $X = \sum_i X_i$. Any particular sequence with k successes and $n-k$ failures has probability $p^k (1-p)^{n-k}$ by independence. There are $\binom{n}{k}$ such sequences, and they are disjoint, so

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The binomial theorem confirms that the masses sum to one:

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + 1-p)^n = 1.$$

Question 12

For $n = 5$, $p = 0.4$, compute $P(X = 0)$, $P(X = 2)$ and $P(X \geq 3)$.

$$P(X = k) = \binom{5}{k} (0.4)^k (0.6)^{5-k}.$$

$$P(X = 0) = (0.6)^5 = 0.07776.$$

$$P(X = 2) = \binom{5}{2} (0.4)^2 (0.6)^3 = 10 \cdot 0.16 \cdot 0.216 = 0.3456.$$

$$P(X = 3) = \binom{5}{3}(0.4)^3(0.6)^2 = 10 \cdot 0.064 \cdot 0.36 = 0.2304,$$

$$P(X = 4) = \binom{5}{4}(0.4)^4(0.6) = 5 \cdot 0.0256 \cdot 0.6 = 0.0768,$$

$$P(X = 5) = (0.4)^5 = 0.01024,$$

$$P(X \geq 3) = 0.2304 + 0.0768 + 0.01024 = 0.31744.$$

Complement check: $P(X = 1) = 5 \cdot 0.4 \cdot (0.6)^4 = 0.2592$, so $P(X \leq 2) = 0.07776 + 0.2592 + 0.3456 = 0.68256$ and $1 - 0.68256 = 0.31744$. The six point masses sum to 1.

Question 13

Show $\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1 - p)$ for $X \sim \text{Binomial}(n, p)$.

Write $X = \sum_{i=1}^n X_i$ with X_i i.i.d. Bernoulli(p). Linearity does not require independence:

$$\mathbb{E}[X] = \sum_i \mathbb{E}[X_i] = np.$$

Independence gives additivity of variance:

$$\text{Var}(X) = \sum_i \text{Var}(X_i) = np(1 - p).$$

(The same identities follow from the PMF via $k \binom{n}{k} = n \binom{n-1}{k-1}$ and $k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2}$; the Bernoulli-sum argument is the one that should be used in a pipeline.) For the numbers of Question 12, $\mathbb{E}[X] = 2$ and $\text{Var}(X) = 1.2$.

Question 14

Write the normal density and explain each parameter.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

- $\mu \in \mathbb{R}$ is the location: mean, median and mode. Translating μ translates the density.
- $\sigma^2 > 0$ is the scale: $\text{Var}(X) = \sigma^2$. Larger σ flattens and widens the bell; the inflection points sit at $\mu \pm \sigma$.
- The prefactor $1/\sqrt{2\pi\sigma^2}$ is the normalising constant that forces $\int f = 1$.

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$. The family is closed under affine transforms: $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

Question 15

For $\mu = 0$, $\sigma^2 = 1$, write the standard normal density and the Z -score.

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in \mathbb{R}.$$

Any $X \sim \mathcal{N}(\mu, \sigma^2)$ standardises by

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Z is the number of standard deviations separating X from μ . Probabilities are then read from the standard CDF Φ : $P(X \leq x) = \Phi((x - \mu)/\sigma)$. Example: $X \sim \mathcal{N}(10, 4)$, $x = 14$ gives $Z = 2$; $x = 7$ gives $Z = -1.5$.

Question 16

Explain the empirical rule (68–95–99.7) for the normal distribution.

For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$P(|X - \mu| \leq \sigma) \approx 0.6827,$$

$$P(|X - \mu| \leq 2\sigma) \approx 0.9545,$$

$$P(|X - \mu| \leq 3\sigma) \approx 0.9973.$$

The exact standard-normal values are $2\Phi(1) - 1$, $2\Phi(2) - 1$ and $2\Phi(3) - 1$. The conventional 95% interval uses 1.96σ rather than 2σ ; 99% uses 2.576σ . Operationally: a residual larger than 3σ under a Gaussian noise model is an outlier candidate; a z -score of 1.96 is the two-sided 5% critical value. The rule is a property of the Gaussian tails, not of every unimodal distribution, and should not be invoked for markedly skewed attributes without a transformation.

Question 17

Explain why the normal appears in measurement errors, PCA, and the central limit theorem.

1. **Measurement errors.** A recorded length, voltage or sensor value is typically the sum of many small, roughly independent disturbances (thermal noise, quantisation, operator variation). The CLT then supplies $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ as the default residual in regression and signal models.
2. **PCA.** A centred Gaussian is completely specified by its covariance. The eigenvectors of that covariance are exactly the principal axes of the concentration ellipsoid $\{x : x^\top \Sigma^{-1} x \leq c\}$. Even when the data are only approximately Gaussian, the same spectral construction remains the variance-maximising orthogonal frame (Question 7), which is why PCA is applied far beyond the Gaussian model that motivates it.
3. **Central limit theorem.** If $X_1, \dots, X_n$ are i.i.d. with finite mean μ and variance σ^2 , then

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Means, estimated coefficients and many other linear statistics are therefore treated as Gaussian for large n , independently of the parent law. That is the structural reason the normal density is the default noise and the default asymptotic approximation in this course.

Question 18

Explain how the covariance matrix leads to an eigenvalue decomposition in PCA.

Stack the centred observations as rows of $\tilde{X} \in \mathbb{R}^{n \times d}$. The (ML) covariance is the Gram matrix of the features,

$$\Sigma = \frac{1}{n} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{d \times d}.$$

Σ is symmetric and positive semi-definite, hence orthogonally diagonalisable:

$$\Sigma = V \Lambda V^\top, \quad V^\top V = I, \quad \Lambda = \text{diag}(\lambda_1 \geq \cdots \geq \lambda_d \geq 0).$$

The j -th column v_j of V is a principal direction; λ_j is the variance of the projected coordinate $z_j = \tilde{X} v_j$. The reconstruction

$$\hat{x} = \bar{x} + \sum_{j=1}^k (v_j^\top (x - \bar{x})) v_j$$

is the orthogonal projection onto the leading invariant subspace and minimises squared reconstruction error among all k -dimensional linear projections (Eckart–Young). The explained-variance ratio of that subspace is $(\lambda_1 + \cdots + \lambda_k) / \text{tr}(\Sigma)$.

Question 19

Explain the relationship between vector projection, the maximum-variance direction, and the largest eigenvalue of the covariance.

Questions 1–4 constructed the projection of a single vector onto a line. PCA asks for the line that maximises the *average* squared projected length of a whole centred sample:

$$\max_{\|u\|=1} \frac{1}{n} \sum_{i=1}^n (u^\top \tilde{x}_i)^2 = \max_{\|u\|=1} u^\top \Sigma u.$$

The constrained stationarity condition $\Sigma u = \lambda u$ is an eigenvalue equation; the maximum value is $\lambda_{\max}(\Sigma)$ and is attained at the corresponding unit eigenvector. Thus:

- the projection formula of Questions 1–3 is applied, not to one b , but to every centred point;
- the scalar α is replaced by the coordinate $u^\top \tilde{x}_i$;
- the line that makes those coordinates as variable as possible is the leading eigen-direction of Σ .

The same identity also shows that a subsequent direction orthogonal to u cannot capture more variance than λ_2 , and so on. Projection, Rayleigh maximisation and the spectral decomposition of Σ are three writings of one fact.

Question 20

Given $x_1 = [1, 2]$, $x_2 = [2, 3]$, $x_3 = [3, 5]$, compute the mean, centre the data, form the covariance, compute its eigenvalues, and interpret the first principal direction.

Mean.

$$\bar{x} = \frac{1}{3}(x_1 + x_2 + x_3) = \begin{pmatrix} 2 \\ 10/3 \end{pmatrix}.$$

Centred sample.

$$\tilde{x}_1 = \begin{pmatrix} -1 \\ -4/3 \end{pmatrix}, \quad \tilde{x}_2 = \begin{pmatrix} 0 \\ -1/3 \end{pmatrix}, \quad \tilde{x}_3 = \begin{pmatrix} 1 \\ 5/3 \end{pmatrix}.$$

Covariance. Using the machine-learning convention $\Sigma = \frac{1}{n} \sum_i \tilde{x}_i \tilde{x}_i^\top$,

$$\sum_i \tilde{x}_i \tilde{x}_i^\top = \begin{pmatrix} 2 & 3 \\ 3 & 14/3 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 2/3 & 1 \\ 1 & 14/9 \end{pmatrix}.$$

(The unbiased estimator with denominator $n - 1 = 2$ is $\frac{3}{2}\Sigma$; eigenvalues simply scale by $3/2$ and the eigenvectors are identical.)

Eigenvalues.

$$\det(\Sigma - \lambda I) = \left(\frac{2}{3} - \lambda\right)\left(\frac{14}{9} - \lambda\right) - 1 = 0 \implies 27\lambda^2 - 60\lambda + 1 = 0,$$

$$\lambda = \frac{10 \pm \sqrt{97}}{9}.$$

$$\lambda_1 = \frac{10 + \sqrt{97}}{9} \approx 2.205, \quad \lambda_2 = \frac{10 - \sqrt{97}}{9} \approx 0.0168.$$

Trace check: $\lambda_1 + \lambda_2 = 20/9 = \text{tr}(\Sigma)$. Almost all of the energy (99.2%) sits in the first component.

First principal direction. $(\Sigma - \lambda_1 I)v = 0$ yields the exact direction

$$v_1 \propto \begin{pmatrix} 9 \\ 4 + \sqrt{97} \end{pmatrix} \approx \begin{pmatrix} 1 \\ 1.539 \end{pmatrix}.$$

The unit vector is approximately $(0.545, 0.838)^\top$. This is the direction of the cloud: the raw points run from $(1, 2)$ to $(3, 5)$, a displacement of $(2, 3)$ whose slope 1.5 matches 1.539 to the precision permitted by three observations. Projecting onto v_1 therefore retains the dominant linear trend; the second component is a small residual orthogonal to that trend (eigenvalue 0.017).

Note

If a marker uses the unbiased covariance $S = \frac{1}{n-1} \sum_i \tilde{x}_i \tilde{x}_i^\top$, the eigenvalues become $\lambda = (10 \pm \sqrt{97})/6 \approx 3.308$ and 0.025, with the *same* first principal direction. State the convention and the direction; do not treat the scale factor as a disagreement about geometry.

End of submission — Assignments I, II and IV